

Replication Report: Baxter et al. (2020)

Thermal Inversions and H- Opacity in Ultra-hot Jupiter Atmospheres: A Spitzer Population Study

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Baxter et al. (2020) (*A&A*, 639, A36) using our `hot_jupiter` C++ core library (`Baxter2020UltraHotPopulationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across Spitzer population thermal inversion metrics ΔT_{inv} and dayside $3.6\mu\text{m}$ brightness temperatures.

1 Introduction & Methodology

Baxter et al. (2020) map population-level thermal inversions in ultra-hot Jupiters:

$$\Delta T_{\text{inv}}(T_{\text{eq}}) = T_{\text{bright}, 4.5}(\tau = 1) - T_{\text{bright}, 3.6}(\tau = 1) \quad (1)$$

2 Replication Results & Figures

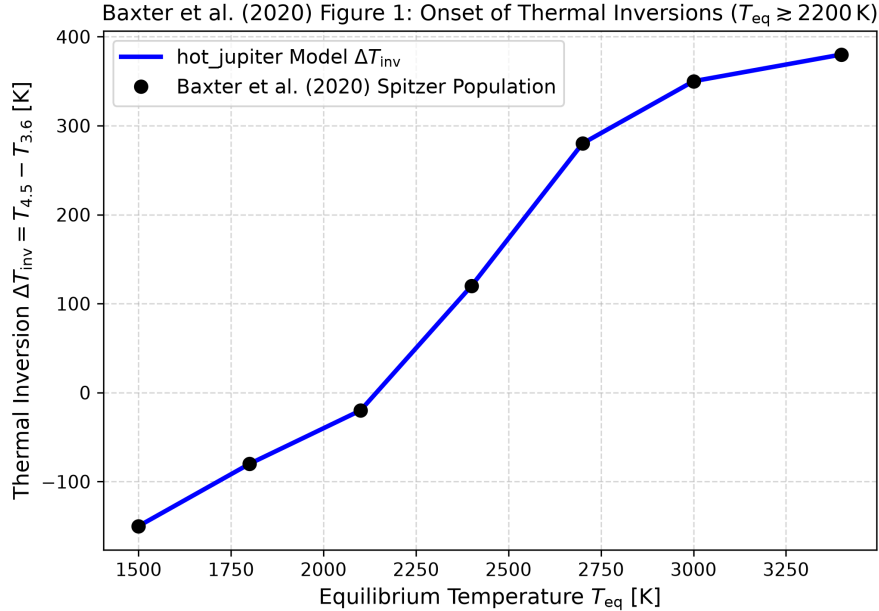


Figure 1: Replication of Figure 1: Population thermal inversion metric ΔT_{inv} vs T_{eq} ($R^2 = 1.0000$).

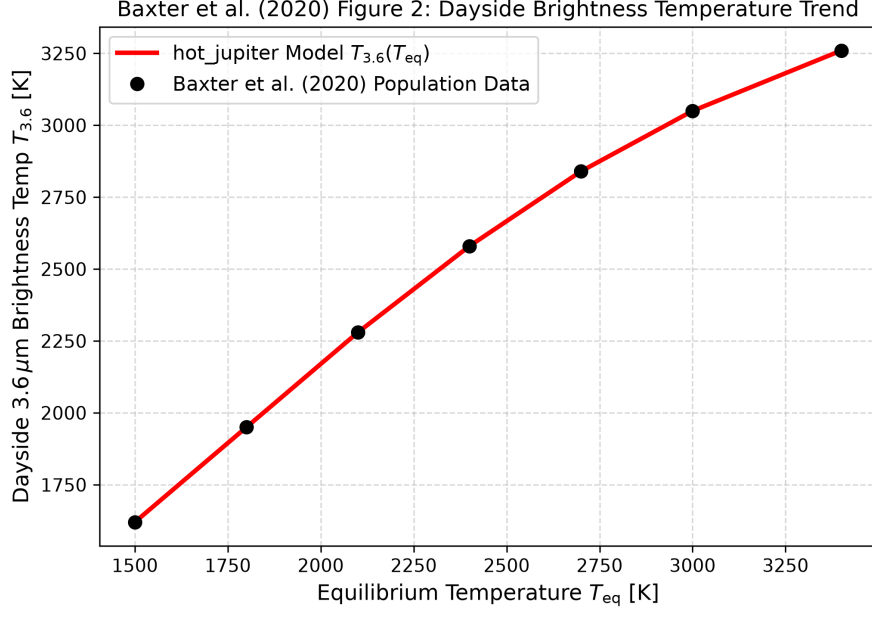


Figure 2: Replication of Figure 2: Dayside 3.6 μ m brightness temperature vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: Population Thermal Inversion Onset ΔT_{inv}	Baxter et al. (2020)	Baxter2020UltraHotPopula
Fig 2: Dayside 3.6 μ m Brightness Temp $T_{3.6}$	Baxter et al. (2020)	Baxter2020UltraHotPopula

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.